

Exact First-Passage Races for Incomparable Subgroup Targets

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Abstract

The frozen C88 receipt gives exact first-passage laws for twenty actual subgroup targets generated by sixteen named labels. We determine the joint race law for every unordered pair that is incomparable in the subgroup poset: 108 pairs, each split into left-first, tie, and right-first outcomes. The certificate resolves every outcome by first-passage rank and by the boundary edges that create it. All counts are exact integers under the scope firewall NO_BAD_EULER_OR_ROOT_NUMBER; no simulation or arithmetic local-data assertion is used.

1 Frozen finite model

Let L be the sixteen named C75 labels and let A_k denote the support of the first k labels in a uniformly random permutation. For each frozen C88 subgroup target H_i define

$$T_i = \min\{k : H_i \leq \Phi(A_k)\}.$$

The C88 receipt supplies the hit indicator of every one of the 2^{16} supports. We restrict to pairs (i, j) for which neither subgroup is contained in the other; the frozen inclusion matrix has exactly 108 such unordered pairs.

2 Race theorem

Theorem 1. *For every unordered incomparable pair (i, j) , the C91 receipt contains exact counts $R_{i,j}^<$, $R_{i,j}^=$, and $R_{i,j}^>$ for $T_i < T_j$, $T_i = T_j$, and $T_j < T_i$, respectively. They satisfy*

$$R_{i,j}^< + R_{i,j}^= + R_{i,j}^> = 16! = 20,922,789,888,000.$$

The receipt also contains the same three counts for each rank $k = 0, \dots, 16$.

Proof. For a support S of size k and a label $x \in S$, consider the boundary edge $S \setminus \{x\} \rightarrow S$. If a target is not hit below the edge and is hit on S , the edge is pivotal. Every rank- k edge represents exactly $(k-1)!(16-k)!$ permutations: the first $k-1$ positions are an ordering of $S \setminus \{x\}$ and the remaining positions are an ordering of its complement. For pair (i, j) , a left-only pivotal edge contributes to $R_{i,j}^<$ precisely when i is first hit on the edge and j is not hit on S ; the symmetric condition gives $R_{i,j}^>$, while simultaneous pivotality gives $R_{i,j}^=$. These three disjoint classes partition every permutation by its first rank at which either target is hit, proving the identity and the rank decomposition. $\square$

3 Certificate inventory

There are 108 pair rows, each with three exact rational probabilities, 17 rank cells, and the corresponding boundary-edge counts. Ninety-nine rows have a nonzero tie class. The independent checker rebuilds all bitset boundary edges directly from C88 and verifies every partition identity. SymPy checks all 324 outcome probability-generating functions and the exact factorial edge weights. Clean replay preserves the evidence digest

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The result is a finite uniform-permutation certificate only. It claims no arithmetic or local data, Euler factors, root numbers, automorphy, full Burnside ring or table of marks, or Hilbert–Polya operator.